FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

FULL TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 10)** contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

2. **Section-C (01 – 10)** contains 10 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

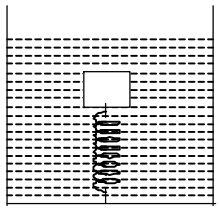
Physics

PART – I

SECTION – A
(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

1. The block shown in the figure is in equilibrium in a liquid and one end of the spring is connected to the base of the vessel. Now the vessel is made to fall freely under gravity. The length of the spring from the equilibrium situation may:



(A) Increase (B) Decrease
(C) Remains constant (D) Data insufficient

2. An electric dipole is placed along x-axis with its centre at origin:
(A) The electric field is perpendicular to each and every point on y-z plane
(B) y-z plane is an equipotential surface
(C) y-z plane is not an equipotential surface
(D) Flux of electric field enclosing dipole is zero

3. A particle of mass *m* makes a head-on elastic collision with a particle of mass 2 m initially at rest. The velocity of the first particle before and after collision are given to be u_1 and v_1. Then which of the following statements is true in respect of this collision?

(A) For all values of u_1, v_1 will always be less than u_1 in magnitude and $|v_1| = \frac{u_1}{3}$.

(B) The fractional loss in kinetic energy of the first particle is $\frac{8}{9}$.

(C) The gain in kinetic energy of the second particle is $\frac{8}{9}th$ of the initial kinetic energy of the first particle

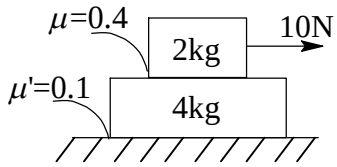
(D) There is a net loss in the energy of the two

Space for Rough work

4. A person of mass 70 kg jump from a 3.0 m height.
   (A) impulse on the man by ground will be 539 N-s
   (B) average force on man if the landing is with stiffed leg and body moves by 1.0 cm during impact is 2.1×10^5 N
   (C) average force on man feet by ground if he land with bent legs and body moves by 50 cm is 4.2×10^3 N
   (D) average force on man feet by ground if he land with bent legs and body moves by 50 cm is $(4.9 \times 10^3$ N)

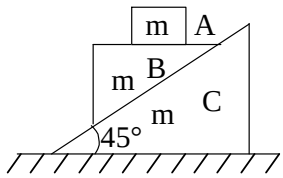
5. Capacitors C_1 = 1 μF and C_2 = 2 μF are separately charged from the same battery. They are allowed to discharge separately through equal resistors
   (A) The currents in the two discharging circuits at t = 0 is zero.
   (B) The currents in the two discharging circuits at t = 0 are equal but not zero.
   (C) The currents in the two discharging circuits at t = 0 are unequal.
   (D) C_1 loses 50% of its initial charge sooner than C_2 loses 50% of its initial charge.

6. For the given two blocks system shown in figure (g=10 m/s^2):



   (A) Acceleration of 2kg block is $1\,m/s^2$
   (B) Acceleration of 4kg block is $2\,m/s^2$
   (C) Friction force between blocks is 8N
   (D) Friction force between lower block and ground is 6N

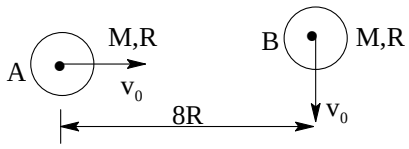
7. The system of block and wedges all of same mass shown in the figure is released from rest. All surfaces are smooth. Then:



   (A) The speed of block A when it falls 50 cm is $2\sqrt{2}\,m/s$
   (B) The speed of block B when A falls 50 cm is $4\,m/s$
   (C) The speed of block B when A falls 50 cm is $\sqrt{10}\,m/s$
   (D) The speed of block C when A falls 50 cm is $\sqrt{2}\,m/s$

Space for Rough work

8. Two spheres are moving as shown. Velocity v_0 is such that spheres are at minimum separation when they touch each other. Then:



(A) $v_0 = \sqrt{\frac{3}{28}\frac{GM}{R}}$

(B) $v_0 = \sqrt{\frac{6}{28}\frac{GM}{R}}$

(C) Speed of sphere A at the time of minimum separation is $\sqrt{\frac{3}{7}\frac{GM}{R}}$

(D) Speed of sphere B at the time of minimum separation is $\sqrt{\frac{6}{7}\frac{GM}{R}}$

9. Two identical fuses are rated at 10A. If they are joined
(A) In parallel, the combination acts as a fuse of rating 20A
(B) In parallel, the combination acts as a fuse of rating 5A
(C) In series, the combination acts as a fuse of rating 10A
(D) In series, the combination acts as a fuse of rating 20A

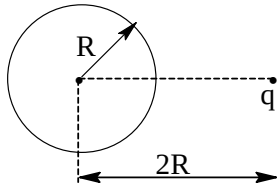
10. Velocity and acceleration vector of a charged particle moving in a magnetic field at some instant are $\vec{v} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = 2\hat{i} + x\hat{j}$. Select the correct alternative (s)
(A) x = -1.5
(B) x = 3
(C) Magnetic field is along z-direction
(D) Kinetic energy of the particle is constant

Space for Rough work

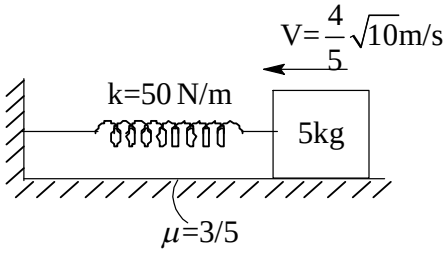
SECTION – C
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

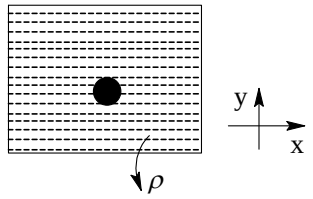
11. A point charge is placed outside a spherical shell as shown in the figure. Find the electric potential (in kilo volt) on the surface of shell (Take: $q=1\mu c$ and $R=50cm$)



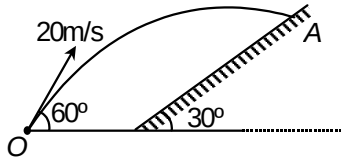
12. A spring block system is placed on a horizontal rough surface as shown in the figure. The block is given velocity when the spring is in natural length. The total distance travelled by the block before it finally comes to rest is $\frac{2}{k}$ meters. Find the value of k



13. A small spherical ball of density same as that of liquid is released from rest in a vessel filled completely with a liquid and accelerating with acceleration $5\hat{i}+5\hat{j}\ m/s$ as shown in the figure. Find the initial acceleration of ball just after the release with respect to vessel.

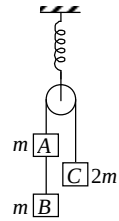


14. A ball is projected from the point *O* with velocity 20 m/s at an angle of $60°$ with horizontal as shown in figure. At highest point of its trajectory it strikes a smooth plane of inclination $30°$ at point *A*. The collision is perfectly inelastic. The maximum height from the ground attained by the ball is $\frac{75}{k}$ meter. Find the value of k? (g = 10 m/s^2)

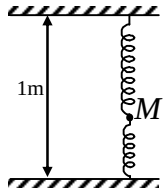


Space for Rough work

15. In the figure shown, pulley and spring are ideal and strings are light and inextensible. Initially all the bodies are at rest when string connecting A and B is cut. Find initial acceleration of C.



16. A small ball of mass $M=1\text{kg}$ is attached to two identical springs of constant 100N/m, which are attached to floor and roof. The springs are initially unstretched. By what distance ball comes down to attain equilibrium? (in cm)

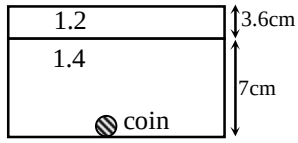


17. In *YDSE*, when a glass plate of refractive index 1.5 of thickness *t* is placed in the path of one of the interfering beams of wavelength λ, intensity at the position where central maximum occurred previously remain unchanged. If the minimum thickness of the glass plate is $k\lambda$. Find the value of k....

18. If the ionization potential of hydrogen atom is 13.6eV, the energy required to remove from the third orbit of hydrogen atom is $k/2\text{eV}$. Find the value of k.....

19. When an object is at a distance 3meters and 1meters from the optical centre of a convex lens, a real and virtual images are formed respectively, with the same magnification. Find the focal length of the lens (in meters)

20. A man observes a coin placed at the bottom of a beaker which contains two immiscible liquids of refractive indices 1.2 and 1.4 as shown in the figure. Find the depth of the coin below the surface, as observed from above.



Space for Rough work

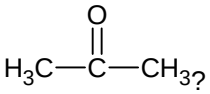
Chemistry

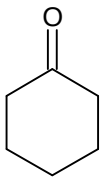
PART – II

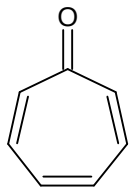
SECTION – A
(One or More than One Options Correct Type)

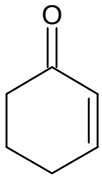
This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

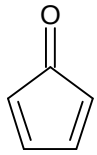
21. Which of the following compound have larger $C-O$ bond length than $C-O$ bond length of



(A) 

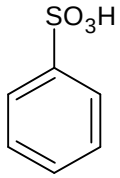
(B) 

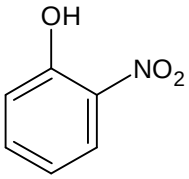
(C) 

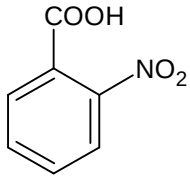
(D) 

22. Which of the following compounds are soluble in $NaHCO_3$?

(A) Picric acid

(B) 

(C) 

(D) 

Space for Rough work

23. Which of the following statement/statements is/are correct?

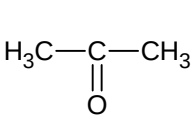
(A) in the $NaCl$ structures, each Na^+ ion is surrounded by $6Cl^-$ ions as nearest neighbours and twelve Na^+ ions as next nearest neighbours.

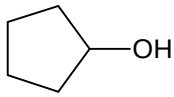
(B) the co-ordination number of cation generally increases as the radius ratio increases

(C) packing efficiencies higher than 0.74 are possible only with cation and anion of different sizes

(D) during a crystal defect electrical neutrality should be maintained.

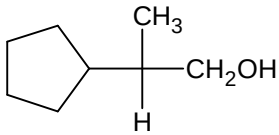
24. An optically active alcohol $A(C_8H_{16}O)$ on oxidation gives B. A on acidic heating gives $C(C_8H_{14})$ as major product. C on ozonolysis produces $D(C_5H_8O)$ and

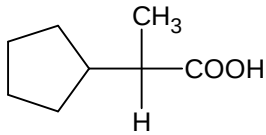


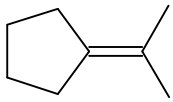
. D on reduction with $LiAlH_4$ gave 

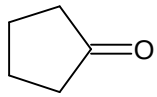
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Identify correct answer:

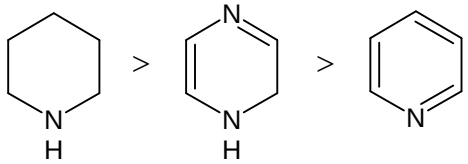
(A) A is 

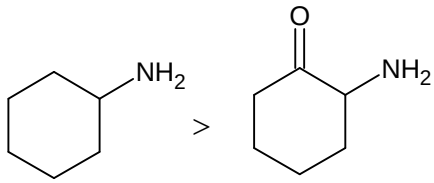
(B) B is 

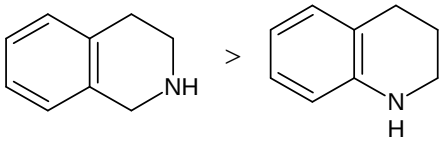
(C) C is 

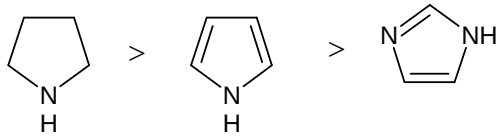
(D) D is 

25. Which of the following order for basic strength is/are correct?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

26. When 5 mL of a 1.0 M HCl solution is mixed with 5 mL of a $0.1M$ $NaOH$ solution, temperature of solution increases by $t°C$. Which of the following(s) can be predicted from this observation:

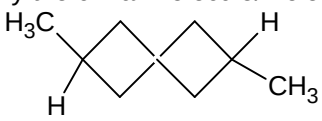
(A) If $5mL$ of $0.1M$ HCl is mixed with 5 mL 0.1 M NH_3 solution, the temperature rise will be less than $t°C$

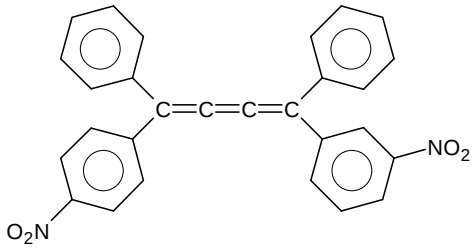
(B) If 5 mL 0.1 M CH_3COOH is mixed with 5 mL 0.1 M $NaOH$, the temperature rise will be less than $t°C$

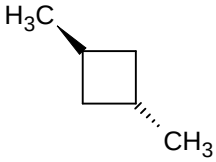
(C) If 10 mL of 1.0 M HCl is mixed with 10 mL of 0.1 M NaOH, the temperature rise will be $2t°C$.

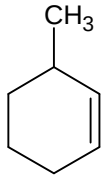
(D) If 10 mL of 0.1 M HCl is mixed with 10 mL of 0.1 M NaCl, the temperature rise will be $t°C$

27. Identify the chiral molecule/molecules:

(A) 

(B) 

(C) 

(D) 

28. Which of the following is/are can produce H_2O_2 on hydrolysis?

(A) BaO_2 (B) PbO_2

(C) MnO_2 (D) Na_2O_2

Space for Rough work

29. The correct statement/statements is/are:
(A) The distance between two octahedral voids of a cubic close packing of atoms with radius r is 2r
(B) The distance between two tetrahedral voids of a cubic close packing of atoms with radius r is $\sqrt{6}\times r$ at the body diagonal axes
(C) The distance between tetrahedral and octahedral void is $\sqrt{\frac{3}{2}}\times r$, where is the radius of atom forming the close packing
(D) The distance between the surfaces of atoms fcc arrangement along an edge is $2\left(\sqrt{2}-1\right)r$

30. Identify the correct relationship:
(A) In $\left[Cu\left(NH_3\right)_4\right]^{2+}$ complex Hybridization of $Cu = dsp^2$
(B) $Fe\left(CO\right)_x$ where x is generally 5
(C) Zn^{2+} Complexes with sp^3d^2 hybridization are diamagnetic in nature
(D) Cr^{3+} Complexes are paramagnetic in nature

SECTION – C
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

31. If four atoms of same radius are placed at the alternate corner of a cube touching each other, than the length of body diagonal of the cube is equal to $\sqrt{x}\times R$, where R is the radius of atom. Find the value of x?

32. The cost of electricity required to deposit 1 g Mg is Rs. 5.00. The cost of 30 g of Al to be deposited is Rs. X. Find the value of $\frac{x}{40}$?

33. A certain buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10}. Find the pH of this buffer solution?

Space for Rough work

34. Co forms dinuclear complex with a sigma bond within two Co atoms. Consider that metal carbonyls follows EAN rule. The complex can be written as $CO_2(CO)_x$. Find the value of x ?

35. Diethyl $-\beta$, $\beta-$ dimethylglutaric acid $(ester)$ is condensed with diethyl oxalate in presence of sodium ethoxide and ethanol to form a major product (P), which on acidic hydrolysis followed by heating gives another product (Q). Calculate total number of C=O bonds in (P) and (Q) compounds?

36. At 298 K, if ΔG_f° of $HCl_{(g)}$ is $1.72\,kJ\,mol^{-1}$, then calculate K_p for the following reversible reaction: $2HCl_{(g)} \rightleftharpoons H_{2(g)} + Cl_{2(g)}$
(Use: at $298\,K : 2.303\,RT = 5700\,J\,mol^{-1}\,and\,\log 2 = 0.30$)

37. Calculate the pH of the following mixture
$50mL$ of $0.05M\ CH_3COOH + 50mL$ of $0.05M\ NH_4OH$
Given : $pK_a = pK_b = 4.74$

38. The dissociation constant of a substituted benzoic acid at $25°C$ is 1×10^{-4}. Find the pH of a $0.01(M)$ solution of its sodium salt.

39. Total no of lone pairs in XeO_3F_2 is $2x$. Find the value of x.

40. The number of water of crystallization in green vitriol is.....

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than One Options Correct Type)

This section contains **10 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** is/are correct.

41. Let $f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{px}{n}\sum\limits_{r=1}^{n}\dfrac{\left[r^2-e^{-x}+r-1\right]}{r(r+1)}\right)+\lambda, & x>0\\ q, & x=0\\ \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{\left\{r^2+r+e^{x}-1\right\}}{r(r+1)}, & x<0\end{cases}$

Is differentiable is R:

$\left([.]\text{is G.I.F. and}\{.\}\text{is F.P. of x}\right)$ Then:

(A) p=1

(B) q=1

(C) p+q+λ=3

(D) if g is inverse of F then $g'(1/2)=2$

42. Consider $f(x)=4x^4-24x^3+31x^2+6x-8$ be a polynomial function and $\alpha,\beta,\gamma,\sigma$ are the roots. $(\alpha<\beta<\gamma<\sigma)$. So:

(A) $\sigma^{p}+\dfrac{1}{\sigma^{\alpha}}+\sigma^{r}+r^{\sigma}=36$

(B) $\sigma^{p}+\dfrac{1}{\sigma^{\alpha}}+\sigma^{r}+r^{\sigma}=18$

(C) $\int\limits_{2\alpha}^{2\beta}\dfrac{x^{\sigma+1}-5x^{r+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=2\ln 2$

(D) $\int\limits_{2\alpha}^{2\beta}\dfrac{x^{\sigma+1}-5x^{r+1}+2\beta|x|+1}{x^2+4\beta|x|+1}dx=\ln 2$

Space for Rough work

43. Each of 2010 boxes in a line contains one red marble and for $1 \le k \le 2010$, the box is the kth position also contain k white marbles. A child begins at the fist box and successively drawn a single marble at random from each box in order. The stops when be fist draws a red marble. Let $p(n)$ be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $p(n) < \frac{1}{2010}$ is:

(A) 44 (B) 45
(C) 46 (D) 47

44. If the normals at four points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are concurrent then $\sum \cos\alpha . \sum \sec\alpha$ is (where $\alpha, \beta, , \gamma, \delta$ are the eccentric angles of the points):

(A) 1 (B) 2
(C) 3 (D) 4

45. Consider the equation $z^2 - (3+i)z + (m+2i) = 0 (m \in R)$. If the equation has exactly one real and one-non-real complex root, then which of the following hold(s) good:

(A) Modulus of the non-real complex root is 2
(B) The value of m is 3
(C) Additive inverse of non-real root is $(-1-i)$
(D) Product of real root and imaginary part of non-real complex root is 2

46. Consider a function $f(x) = \sin^{-1}\frac{2x}{1+x^2} + \cos^{-1}\frac{1-x^2}{1+x^2} + \tan^{-1}\frac{2x}{1-x^2} - a\tan^{-1}x (a \in R)$, the value of a if $f(x) = 0$ for all x:

(A) 6 (B) -6
(C) 2 (D) -2

Space for Rough work

47. Solution of differential equation $f'(x)\,f'''(x)=3\left(f''(x)\right)^2, \left(y=f(x)\right)$ is:

(A) $x=k_1y^2+k_2y+k_3$ (B) $y=k_1x^2+k_2+k_3$

(C) $x=\sin y$ (D) None of these

48. $f(x)=\begin{cases} e^{-\sqrt{|\ln\{x\}|}}-\{x\}\sqrt{\frac{1}{|\ln\{x\}|}} \\ 0 \qquad \text{otherwise}\end{cases}$ Where ever it is defined is:

(A) not a periodic function
(B) even function
(C) range of $f(x)$ contains more than 1 element
(D) None of these

49. $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\frac{m^2n}{\left(n.3^m+m.3^n\right)}=\frac{p}{q}$ (Where p & q are coprime) then:

(A) $p+q=41$ (B) $p<q$

(C) p is perfect square (D) q is perfect square

50. If $f(x)=\sum_{r=0}^{20}{}^{20}C_r\left(\ln(1+x)\right)^r, g(x)=\sum_{r=0}^{20}(-1)^r\,{}^{20}C_r\left(1+3\sin x\right)^{20-r}$ then:

(A) $\sum_{r=0}^{\infty}\left(g(\pi/6)\right)^r=\frac{6^{20}}{2^{20}-1}$

(B) $\sum_{r=0}^{\infty}\left(g(\pi/6)\right)^r=\frac{6^{30}}{2^{20}-1}$

(C) If $h(x)=\left[f(x)\right]^{1/20}$ *then* $\sum_{r=0}^{\infty}\left[h\left(x^{2^r}\right)-1\right](where\,|x|<1)=-\ln(1-x)$

(D) $h(x)=\left[f(x)\right]^{1/20}$ *then* $\sum_{r=0}^{\infty}\left[h\left(x^{2^r}\right)-1\right](where\,|x|<1)=\ln(1-x)$

Space for Rough work

SECTION – C
(One Integer Value Correct Type)

This section contains **10 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

51. Let $p(x)$ be a polynomial of degree 6 with leading coefficient unity and $p(-x)=p(x)\forall x\in R$. Also $\left(p(1)+3\right)^2+p^2(2)+\left(p(3)-5\right)^2=0$ then $\sqrt{\text{-4-p}(0)}$ is....

52. $\lim\limits_{n\to\infty}\left[\left((2009)^{2010}\right)^n+\left((2010)^{2009}\right)^n\right]^{1/n}$ is equal to $\left|a^b\right|$ where $a,b\in N$ then $a-b$ is....

53. The value of x, where $f(x)=\sqrt{x+\sqrt{8x-16}}+\sqrt{x-\sqrt{8x-16}}$ is not differentiable, is....

54. A and B are two square matrices such that $A^2B=BA$ and if $(AB)^{10}=A^kB^{10}$ then the value of $k-1020$ is....

55. $AI_1I_2I_3$ is an excentral triangle of equilateral triangle ΔABC such that $I_1I_2=4$ unit, if ΔDEF is pedal triangle of ΔABC then $\sqrt{\dfrac{Ar(\Delta I_1I_2I_3)}{Ar(\Delta DEF)}}=$

56. If two points P and Q on $9x^2-5y^2=1$ (whose centre is C) be such that CP is perpendicular to CQ then value of $\dfrac{1}{\text{CP}^2}+\dfrac{1}{\text{CQ}^2}$ is.....

57. If the equation $x^3-3x+1=0$ has three roots x_1,x_2,x_3 where $x_1<x_2<x_3$ then the value of $\left(\{x_1\}+\{x_2\}+\{x_3\}\right)$ is $\left(\{.\}\text{F.P.of x}\right)$

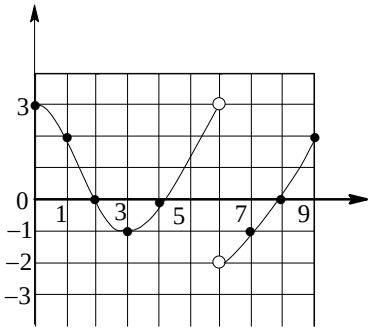
Space for Rough work

58. Let $S=\{(x,y);x^2+y^2-6x-8y+21\le 0\}$ then

$$\max\left\{\frac{12x}{7}-\frac{5y}{7};(x,y)\in S\right\}+\min\left\{\frac{1}{2}(x^2+y^2+1)+(x-y);(x-y)\in S\right\}$$

$$-\min\left\{\frac{\sqrt{3}y+|x-3|}{|x-3|};(x,y)\in S\right\}$$

59. The graph of the derivative f of a continous function f is shown with f(0) = 0



If $f(x)$ has local minima at $x=x_1$ and $x=x_2$; f has inflection point at $x=k$; number of critical points of $y=f(x)$ is w then x_1+x_2-k-w is.....

60. If $f(x)=\int_{-2}^{x}\left(t^4-bt^3+(b+1)t^2-bt+b\right)dt$ strictly increases $\forall x\in R$ then no. of is integers in range of b is.

Space for Rough work